

Machine Learning Midterm Report

- **Introduction:**

Machine Learning is a field of Artificial Intelligence that helps computers learn from data and make predictions. During the first four weeks, I learned the basic concepts needed to start working with Machine Learning. The course began with Python programming and gradually moved towards data handling, data analysis, regression models, and classification models. This report summarizes the important concepts and skills I learned during these four weeks.

- **Week 1: Python Fundamental**

In the first week, I learned the basics of Python programming. I studied variables, data types, operators, input and output statements, and simple calculations. I also learned how to use conditional statements such as if-else and loops such as for and while loops.

Another important topic was Python data structures, including lists, tuples, sets, and dictionaries. These help in storing and organizing data efficiently. I also learned how to create and use functions, which make programs easier to read and reuse.

By the end of the week, I was able to write basic Python programs and solve simple problems. This week helped me build the programming skills required for Machine Learning.

- **Week 2: Data Handling and Exploratory Data Analysis**

In the second week, I learned about some important Python libraries used in Data Science and Machine Learning. These libraries were NumPy, Pandas, and Matplotlib.

Using NumPy, I learned how to work with arrays and perform mathematical operations on data. I also learned how to reshape arrays and perform calculations more efficiently.

Using Pandas, I learned how to work with datasets using DataFrames. I practiced loading CSV files, selecting rows and columns, filtering data, handling missing values, and cleaning datasets. I understood that data cleaning is an important step before building any machine learning model.

Using Matplotlib, I learned how to create different types of graphs such as line plots, bar charts, histograms, and scatter plots. These graphs helped me understand the data better and find patterns in it.

I also learned Exploratory Data Analysis (EDA), where I studied how to inspect datasets, check summary statistics, find missing values, and understand the overall structure of data before using it in Machine Learning.

- **Week 3: Linear Regression and Model Evaluation**

In Week 3, I learned about Linear Regression, which is one of the simplest and most important machine learning algorithms. It is used to predict numerical values by finding a relationship between input data and output data.

I learned how a regression model finds the best fitting line for a dataset. I also studied Mean Squared Error (MSE), which measures the difference between actual values and predicted values. A smaller MSE means the model is making better predictions.

Another important topic was the R^2 Score. This score helps us understand how well the model explains the data. A value closer to 1 means the model is performing well.

I learned why datasets are divided into training and testing sets. The training set is used to teach the model, while the testing set is used to check how well the model works on new data.

I also learned Ridge Regression and LASSO Regression. These methods help improve the performance of a model and reduce unnecessary complexity. LASSO Regression can also help identify less important features in a dataset.

In addition, I learned how to create graphs such as Actual vs Predicted plots and Residual plots to understand model performance visually. I implemented Linear Regression, Ridge Regression, and LASSO Regression using Scikit-Learn and compared their results using different train-test splits.

• **Week 4: Classification and Logistic Regression**

In Week 4, I learned about classification problems. Unlike regression, where the output is a number, classification predicts categories such as Yes/No, Spam/Not Spam, or Disease/No Disease.

I learned why Linear Regression is not suitable for classification problems and how Logistic Regression solves this issue. Logistic Regression uses a Sigmoid Function to convert values into probabilities between 0 and 1.

I learned how probabilities can be used to decide which class a data point belongs to. I also studied the concept of log-odds and how Logistic Regression uses them to make predictions.

Another important topic was binary classification, where there are only two possible classes. I also learned about multiclass classification, where there are more than two classes. Different methods such as One-vs-Rest (OvR), One-vs-One, and Softmax Regression were introduced for handling such problems.

I learned about decision boundaries, which separate different classes, and classification thresholds, which affect how predictions are made. I also learned how prediction probabilities can show the confidence of a model's prediction.

Using Scikit-Learn, I implemented Logistic Regression models and used functions such as `predict()` and `predict_proba()` to make predictions and view probabilities.

Conclusion:

During the first four weeks of this Machine Learning course, I learned the complete basic workflow of Machine Learning. I started with Python programming and then moved on to data handling, data cleaning, visualization, regression models, model evaluation, and classification techniques.

I gained practical experience using NumPy, Pandas, Matplotlib, and Scikit-Learn. I learned how to prepare data, build machine learning models, evaluate their performance, and understand their results.

These four weeks have given me a strong foundation in Machine Learning and prepared me to learn more advanced topics in the coming weeks.